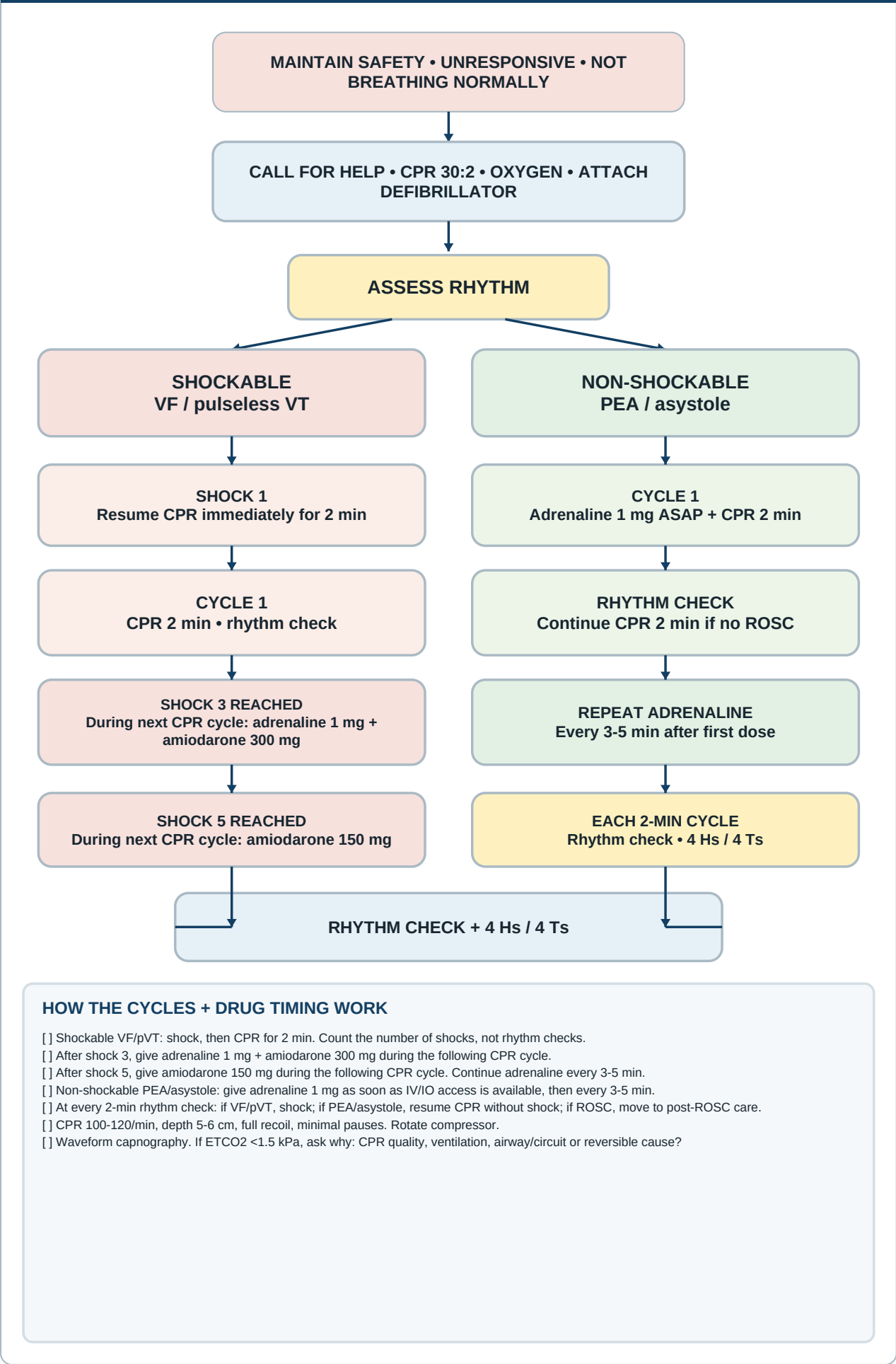


ADULT ADVANCED LIFE SUPPORT - ONE-PAGE COGNITIVE AID

UK paramedic practice | Core 2-minute cycle, exact arrest medicines, reversible causes, special circumstances and ROSC

1. CORE ALS ALGORITHM - CONTINUE 2-MINUTE CYCLES



2. REVERSIBLE CAUSES - TICK + TREAT

- [] Hypoxia: open airway; oxygenate and ventilate.
- [] Hypovolaemia: control haemorrhage/loss; replace volume.
- [] Hyper/hypokalaemia + metabolic: test and treat.
- [] Hypo/hyperthermia: measure core temperature; correct.
- [] Thrombosis: coronary or pulmonary pathway.
- [] Tension pneumothorax: immediate decompression.
- [] Toxins: antidote/supportive care; seek expert advice.
- [] Tamponade: trauma/specialist pathway.

3. SPECIAL CIRCUMSTANCES - WHAT CHANGES?

- WITNESSED, MONITORED VF/pVT: STACKED SHOCKS**
- Only when VF/pVT arrest or re-arrest is witnessed and monitored by the crew, with a defibrillator immediately available.
 - Up to 3 stacked shocks may be considered. Check rhythm after each shock; stop if ROSC or rhythm changes.
 - For adrenaline/amiodarone timing, the whole initial stacked sequence counts as shock 1, not shocks 1-3.
- HYPOTHERMIA: KEEP 2-MIN CYCLES**
- <30°C: continue CPR/rhythm checks; maximum 3 shocks, then defer further shocks until >30°C.
 - <30°C: withhold adrenaline. If shockable, one amiodarone 300 mg loading dose may be given; defer further doses.
 - 30-35°C: resume shocks; adrenaline every 6-10 min, not every alternate cycle.
 - >35°C: standard 3-5 min drug timing. Prevent heat loss, rewarm and seek ECPR early.
 - Target normocapnia; hypothermia may require lower minute ventilation.
- TORSADES DE POINTES**
- Pulse: magnesium sulfate 2 g IV over 10 min; correct K⁺/Mg²⁺; seek expert help.
 - Pulseless: treat as VF/pVT; defibrillate and give magnesium 2 g IV/IO.
- PREGNANCY >20 WEEKS**
- Manual left uterine displacement; standard shocks and ALS drugs.
 - Early obstetric/surgical/neonatal help; prepare resuscitative hysterotomy as soon as possible.
- DROWNING / TRAUMA / ANAPHYLAXIS**
- Drowning: 5 initial breaths with 100% oxygen; apply hypothermia changes if cold.
 - Trauma: treat haemorrhage, hypoxia, bilateral tension pneumothorax and tamponade in parallel.
 - Anaphylaxis before arrest: IM adrenaline + IV crystalloid; in arrest use standard ALS adrenaline.
- TOXINS: NALOXONE**
- Opioid respiratory depression WITH a pulse: ventilate first.
 - Naloxone 400 mcg IV/IM; if no response after 1 min give 800 mcg, repeat 800 mcg, then 2 mg; reassess diagnosis and monitor re-sedation.
 - Confirmed cardiac arrest: immediate CPR/standard ALS; naloxone is not routine.

4. MEDICINES - DOSE, VOLUME, TIMING

- OXYGEN**
- Highest available concentration during CPR.
 - After ROSC, titrate to SpO₂ 94-98%.
- ADRENALINE 1:10,000 (100 mcg/mL)**
- 1 mg = 10 mL IV/IO.
 - PEA/asystole: give ASAP.
 - VF/pVT: give after shock 3.
 - Repeat every 3-5 min after first dose.
- AMIODARONE 30 mg/mL**
- After shock 3: 300 mg = 10 mL IV/IO.
 - After shock 5: 150 mg = 5 mL IV/IO.
- MAGNESIUM SULFATE**
- Torsades: 2 g IV over 10 min with pulse; 2 g IV/IO if pulseless.
- SODIUM CHLORIDE 0.9%**
- 20 mL IV/IO drug flush. Fluid bolus only for a defined indication; reassess.
- OTHER INDICATIONS - NOT ROUTINE ARREST DRUGS**
- Glucagon: severe hypoglycaemia 1 mg IM; selected beta-blocker toxicity requires expert advice.
 - Hydrocortisone: suspected adrenal crisis 100 mg IV/IM.
 - Atropine: symptomatic bradycardia 500 mcg IV; repeat every 3-5 min to 3 mg maximum.

5. ROSC: CONFIRM, STABILISE, TRANSFER

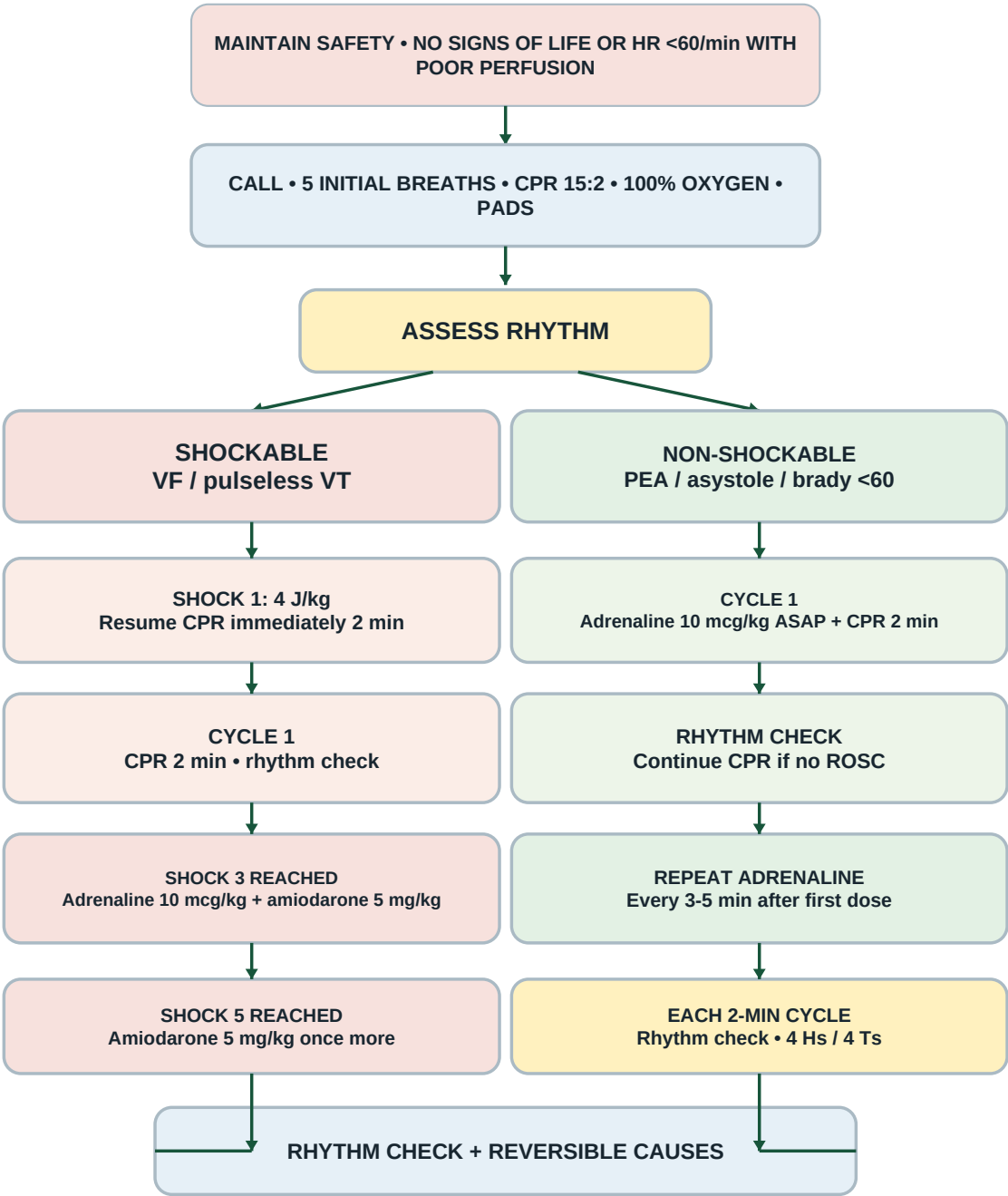
- Confirm signs of circulation, then stop CPR and perform ABCDE.
- Maintain airway and waveform capnography; aim normal PaCO₂.
- Titrate oxygen to SpO₂ 94-98%.
- Treat hypotension; target SBP >100 mmHg.
- Obtain 12-lead ECG; identify and treat the cause.
- Control temperature; check glucose and treat seizures.
- Pre-alert and transfer to the appropriate receiving centre; anticipate re-arrest.

- CONSIDER / ESCALATE**
- Coronary angiography / PCI when indicated.
 - Mechanical CPR for transfer/treatment when appropriate.
 - Ultrasound for reversible causes without interrupting CPR.
 - Extracorporeal CPR where available and indicated.
 - After 3 failed shocks, consider antero-posterior pad position.

PAEDIATRIC ADVANCED LIFE SUPPORT - ONE-PAGE COGNITIVE AID

UK paramedic practice | Core 2-minute cycles, weight-based medicines, reversible causes, special circumstances and ROSC

1. CORE PALS ALGORITHM - CONTINUE 2-MINUTE CYCLES



HOW THE CYCLES + DRUG TIMING WORK

- [] VF/pVT: shock 4 J/kg, then CPR 2 min. Count shocks, not rhythm checks.
- [] After shock 3: adrenaline 10 mcg/kg + amiodarone 5 mg/kg during CPR.
- [] After shock 5: repeat amiodarone 5 mg/kg once. Continue adrenaline every 3-5 min.
- [] PEA/asystole/brady <60 with poor perfusion: adrenaline 10 mcg/kg ASAP, then every 3-5 min.
- [] At each rhythm check: shock VF/pVT; do not shock PEA/asystole; if ROSC, start post-ROSC care.
- [] From shock 5 in refractory VF/pVT: escalate stepwise to 8 J/kg, maximum 360 J.
- [] If ETCO2 <1.5 kPa, ask why: CPR quality, ventilation, airway/circuit or reversible cause?

2. REVERSIBLE CAUSES - TICK + TREAT

- [] Hypoxia: open airway; oxygenate and ventilate.
- [] Hypovolaemia: control loss; 10 mL/kg fluid if indicated, reassess.
- [] Hyper/hypokalaemia + metabolic: test and treat.
- [] Hypo/hyperthermia: measure core temperature; correct.
- [] Thrombosis: coronary or pulmonary pathway.
- [] Tension pneumothorax: immediate decompression.
- [] Toxins: antidote/supportive care; seek expert advice.
- [] Tamponade: trauma/specialist pathway.

3. SPECIAL CIRCUMSTANCES - WHAT CHANGES?

- HYPOTHERMIA: KEEP 2-MIN CYCLES**
- <30°C: continue CPR/rhythm checks; maximum 3 shocks, then defer further shocks until >30°C.
 - <30°C: withhold adrenaline and defer further antiarrhythmic doses.
 - 30-35°C: resume shocks; adrenaline every 6-10 min.
 - >35°C: standard 3-5 min timing. Prevent heat loss, rewarm and seek specialist/ECPR advice.
 - Target normocapnia; avoid hyperventilation.
- TORSADES DE POINTES**
- Pulse: magnesium sulfate 25-50 mg/kg IV over 10-15 min; max 2 g; may repeat once.
 - Pulseless: treat as VF/pVT and defibrillate.
- DROWNING / TRAUMA / ANAPHYLAXIS**
- Drowning: 5 initial breaths with 100% oxygen; apply hypothermia changes if cold.
 - Trauma: treat haemorrhage, hypoxia, bilateral tension pneumothorax and tamponade in parallel.
 - Anaphylaxis before arrest: IM adrenaline by age + crystalloid; in arrest use PALS adrenaline.
- PREGNANT ADOLESCENT**
- Use adult pregnancy modification: manual left uterine displacement if >20 weeks.
 - Standard shocks/drugs; call obstetric, paediatric and neonatal support early.
- TOXINS: NALOXONE**
- Opioid respiratory depression WITH a pulse: ventilate first; use current JRCALC/BNFc age-weight naloxone dose.
 - Confirmed arrest: immediate CPR/standard PALS; naloxone is not routine.

4. WEIGHT-BASED MEDICINES + SHOCKS

- OXYGEN**
- 100% during CPR; after ROSC titrate to SpO2 94-98%.
- ADRENALINE 1:10,000 (100 mcg/mL)**
- 10 mcg/kg = 0.1 mL/kg IV/IO.
 - PEA/asystole/brady <60: give ASAP.
 - VF/pVT: give after shock 3.
 - Repeat every 3-5 min; maximum single dose 1 mg.
- AMIODARONE 30 mg/mL**
- 5 mg/kg = 0.167 mL/kg IV/IO after shocks 3 and 5 only.
 - Maximum 300 mg after shock 3; 150 mg after shock 5.
- FLUID / GLUCOSE**
- Balanced isotonic crystalloid or NaCl 0.9%: 10 mL/kg IV/IO if indicated; reassess.
 - Known hypoglycaemia: 10% glucose 2 mL/kg IV/IO; recheck 5-10 min.
- OTHER INDICATIONS - NOT ROUTINE ARREST DRUGS**
- Glucagon severe hypoglycaemia: >=25 kg or >8 y 1 mg IM; <25 kg or <8 y 0.5 mg IM.
 - Hydrocortisone adrenal crisis: <1 y 25 mg; 1-5 y 50 mg; >=6 y 100 mg IV/IM.
 - Atropine: 20 mcg/kg IV/IO for specified bradycardia indications; max 600 mcg.
- Approximate age weights. Amiodarone = 5 mg/kg; glucose 10% = 2 mL/kg (*50 mL cap shown). Verify actual weight/current RCUK + JRCALC charts.

Age	Wt	Adr. mL	Amio mg	Fluid mL	Glucose mL	Glucagon	Hydrocort.	Shock J
<1 mo	3.5 kg	0.35	17.5	35	7	0.5 mg	25 mg	14
1 mo	4 kg	0.4	20	40	8	0.5 mg	25 mg	16
3 mo	5 kg	0.5	25	50	10	0.5 mg	25 mg	20
6 mo	7 kg	0.7	35	70	14	0.5 mg	25 mg	28
1 y	10 kg	1.0	50	100	20	0.5 mg	50 mg	40
2 y	12 kg	1.2	60	120	24	0.5 mg	50 mg	48
4 y	16 kg	1.6	80	160	32	0.5 mg	50 mg	64
6 y	20 kg	2.0	100	200	40	0.5 mg	100 mg	80
8 y	26 kg	2.6	130	260	50*	1 mg	100 mg	104
10 y	30 kg	3.0	150	300	50*	1 mg	100 mg	120
12 y	38 kg	3.8	190	380	50*	1 mg	100 mg	152
14 y+	50 kg	5.0	250	500	50*	1 mg	100 mg	200

5. DURING CPR + ROSC

- [] CPR 15:2; correct rate, depth and full recoil; minimise pauses; rotate compressor.
- [] Advanced airway ventilation: infant 25/min; 1-8 y 20/min; 8-12 y 15/min; >12 y 10/min.
- [] IV/IO access; waveform capnography; check glucose and temperature.

WHEN ROSC IS ACHIEVED

- Confirm circulation, stop CPR, perform ABCDE.
- SpO2 94-98%; normal PaCO2; maintain systolic and mean BP above age 10th percentile.
- Treat cause; glucose control; avoid/manage hypothermia.
- Pre-alert appropriate paediatric receiving team; anticipate re-arrest.
- Provide family support and follow local safeguarding/documentation policy.